

GABRIELE NICOLA NAPOLI

PhD in “Earth Processes And Management Of Resources And Risks For A Resilient Society And Territory”

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ACADEMIC EXPERIENCE

PhD in Geosciences (National Interest Announcement - University of Bari Aldo Moro)

University of Naples

October 2023 – Current Naples, Italy

Fellowship Title: Evaluation based on a multi-risk test bed, taking into account the interaction between natural disasters: towards a holistic definition of mitigation measures.

- Risk Assessment: Developed methods for predicting risks associated with natural processes threatening human life;
- Disaster Interaction Analysis: Studied the compounded impact of interacting natural disasters, which can exceed the sum of individual threats;
- Test Bed Utilization: Employed test beds to represent the complexity of environmental systems and their potential impact on anthropized systems;
- Multi-risk Assessment: Used test beds for multi-risk assessment, considering interactions between multiple geological/natural threats;
- Impact Analysis: Conducted consistent assessments of the impact on exposed systems, such as urban settlements and infrastructures;
- Procedure Development: Developed a new procedure for the assessment and management of multi-hazard risk based on the use of test beds;
- Holistic Approach: Incorporated both the interaction between threats and the need for a new holistic criterion for selecting mitigation measures into the new procedure

GIS Analyst, Remote Sensing Scientist/Technician (Erasmus Trainee)

University of Debrecen

August 2020 – November 2020 Debrecen, Hungary

- PyQGIS (Python environment of QGIS)
- LiDAR point cloud analysis and DTM generation in fluvial environment;
- Heavy metal accumulation of fluvial forms;
- Merphonetic analysis for object detection in floodplain;
- Digitization of Local Climate Zones (LCZ) scheme;
- Vegetation classification by Landsat 8, Sentinel 2, PlanetScope and SkySat satellite images;
- Image classification by the ENVI program

IN A NUTSHELL

‘ PhD student in Geological Sciences and Technologies with a passion for Remote Sensing, GIS, and Spatial Data Science. My skills include describing geomorphological phenomena using various GIS-related tools and software. I also have a keen interest in programming languages and computer technologies applied to geosciences. ’

STRENGTHS

- O.S.: Linux-based, Windows
- Text processors (LaTeX, MSOffice, LibreOffice)
- R, Matlab, CSS, JavaScript, Python
- 3D point cloud processing
- AutoCAD Map 3D WebGIS
- Imagery Processing

LANGUAGES

- Italian ● ● ● ● ●
- English ● ● ● ● ●
- German ● ● ● ● ●
- Spanish ● ● ● ● ●

SEMINARS

- 10th Hungarian GIS Conference and Exhibition, **University of Debrecen** (Debrecen, Hungary, 23 – 24 May 2019).
- Geomorphometry based on different Digital Terrain Models, **University of Debrecen** (Debrecen, Hungary, 23-24 May 2019).
- 3rd International Symposium of Soil and Water Bioengineering: Materials and Techniques in Ecological Restoration (S.Agata di Militello, Italy, 13 – 15 October 2017).
- "Terza Edizione Settimana del Pianeta Terra: Geositi carsici della Sicilia" (Alcara Li Fusi, Italy, 17 – 18 October 2015).

Environmental Geologist, GIS Analyst (Curricular Traineeship)

Nebrodi Park Authority (ITA: 'Ente Parco dei Nebrodi')

📅 December 2019 – January 2020 📍 S.Agata di Militello, Sicily, Italy

- Study and analysis of the most important geosites within the territory of Nebrodi Park. Use of GIS Software (QGIS) for the creation of charts, mainly used for tourism purposes. Historical outline of the term 'geosite' and its legislation. Description of national and regional catalogs. Geological and geomorphological overview of the geosites examined, including point cloud(s).

GIS Analyst, Remote Sensing Scientist/Technician (Erasmus Student)

University of Debrecen

📅 March 2019 – June 2019 📍 Debrecen, Hungary

- Use of GIS software (QGIS, SAGA GIS, ESRI software packages).
- Use of UAV (Unnamed Aerial Vehicles), LiDAR, Nikon and Canon DSLR Cameras for imagery capture.
- Use of image processing software (CloudCompare Stereo and Agisoft Metashape Professional), for the creation of Point Cloud(s).

Environmental Geologist, GIS Analyst (Curricular Traineeship)

Ente Parco dei Nebrodi

📅 August 2015 – October 2015 📍 S.Agata di Militello, Sicily, Italy

Use and analysis of 'Italian Hydrogeological Hazard' charts (PAI; ITA: 'Piano stralcio di bacino per l'Assetto Idrogeologico') and their official reports. Analysis of landslides with the highest probability of being triggered. Use of GIS software (QGIS and ESRI's 'ArcMap').

PROJECTS

Cloud-based interactive susceptibility modeling of natural hazards in Google Earth Engine

University of Palermo (Italy), University of Twente (The Netherlands), University of Bologna (Italy)

📅 Remote 📍 2021 – 2022

Creation of a tool (STGEE) for susceptibility estimates for gully erosion occurrences in a specific study area: i) Creation of Slope Units; ii) Processing of slope units with functions ('gridcoll') that allow the aggregation of predictor information in space and time, over a chosen mapping unit; iii) Use of a Random Forest Classifier to distinguish locations prone to gully erosion from locations where this process is not likely to occur; iv) Use of visualization tools.

High resolution monitoring of river channel bathymetry, flow velocities and bank erosion of Sajó River, Hungary

University of Debrecen (Hungary)

📅 Debrecen (Hungary), Remote 📍 2019 – 2020

i) Geological and geomorphological framework of the study area; ii) Collection of digital data via DSLR camera and Terrestrial Laser Scanner (TLS); iii) Photogrammetric processing of image acquisitions and previous survey, using a photogrammetric processing software; Reconstruction of point clouds using the Structure-from-Motion

EDUCATION

M.Sc. in Geological Sciences and Technologies

University of Palermo

📅 2017 – 2020 📍 Palermo, Italy

Thesis title: "Comparison Of Different Surface Reconstruction Methods For The Quantification Of Bank Erosion", Hungary

B.Sc. in Geological Sciences

University of Palermo

📅 2012 – 2017 📍 Palermo, Italy

Thesis title: "Landslides in River Systems", Italy

VOLUNTEERING

OpenStreetMap Team

Worldwide

📅 Remote 📍 April 2022 – Current

Disaster and Humanitarian Relief

Associate URISA's GISCorps

Worldwide

📅 Remote 📍 April 2022 – Current

- Disaster and Humanitarian Relief

Servizio Civile Universale

Acquedolci, Province of Messina, Italy

📅 Blended 📍 July 2021 – Oct. 2021

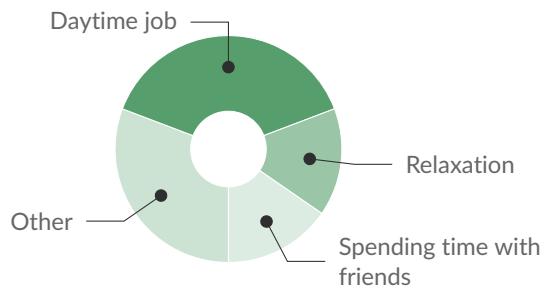
Project: Riqualficazione e Salvaguardia Ambientale

ENG - The goal of the program is to enhance the quality and quantity of environmental interventions, with the aim of promoting environmental issues. This is achieved through initiatives directed at citizens and implemented in areas where the program will be active.

ITA - L'obiettivo del programma è di migliorare la qualità e la quantità di interventi in favore dell'ambiente al fine di promuovere la sensibilità verso i temi ambientali, attraverso iniziative specifiche nei confronti dei cittadini presenti sul territorio sul quale verrà attuato il programma.

(SfM) technique; iv) Use of the CloudCompare software for the comparison of the point clouds; v) Application of the C2C and M3C2 algorithms to compare the point clouds obtained with SfM and TLS, and evaluation of the accuracy of the models.

A DAY OF MY LIFE



PUBLICATIONS

Journal Articles

- Titti, G., Napoli, G., Conoscenti, C., & Lombardo, L. (2022). Cloud-based interactive susceptibility modeling of natural hazards in google earth engine. doi:10.31223/X5SW6S

Summer Schools, Conferences and Workshops

- AIGAA - Italian Association for Applied and Environmental Geology. (n.d.). Complex of SS. Marcellino e Festo, Largo S. Marcellino 10, Naples, 27-29 June (2024).
- International Association for Engineering Geology and the Environment (IAEG), THIRD SUMMER SCHOOL: Natural hazards, societal impact and risk from theory to practice. (n.d.). University of Valle D'Aosta, 1-9 July (2024).
- LARAM (Landslide Risk Assessment and Mitigation) School 2024, Geotechnical Engineering Group (GEG), University of Salerno. (n.d.). Palazzo della Provincia, Salerno, 9-20 September 2024.
- XVIII National Conference of the "GIT – Geosciences and Information Technologies" and "SI – Hydrogeology Section, 16-19 June 2024". (n.d.). Malnisio di Montereale Valcellina (PN).
- Designing nudges for risk mitigation, Workshop Return (TS3 – WP5 – Task 5.4). (2024). AULA BALAB, ex Palazzo delle Poste, Bari, (BA), 22-23 April 2024.

Conference Proceedings

- G.N., N., & G., T. (2024). Cloud-based natural hazard susceptibility mapping with google earth engine. In *Xviii convegno nazionale delle sezioni "git – geosciences and information technologies" e "si – sezione di idrogeologia"*, Antica Centrale idroelettrica 'A. Pitter', Malnisio (PN).
- Napoli G.N., C. C., & L., B. (2021). Comparison of different surface reconstruction methods for the quantification of bank erosion. In *1st international joint congress on "sustainable management of cultural landscapes in the context of the european green deal"*, Santo Stefano di Camastra (Italy).

REFERENCE CONTACTS

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Prof. Maria Polese

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PRIVACY DATA AND SIGNATURE

I give consent to process my data with the purpose of the recruitment process, in accordance with the Regulation of the European Parliament 679/2016, regarding the protection of natural persons and free movement of such data.

June 12, 2026

Nyola Gabriele Nido